



ECB Communication Intelligence Quantile Hawkishness & Bund Reaction

Lagarde Tone Barometer | Quantile Regression | t-SNE Market Mapping
Bayesian Change-Point Detection | Scenario Analysis

Quantitative Research | June 11, 2026

1. Hawkish/Dovish Lexicon Construction

We build a 70-term scored lexicon seeded from the ECB June 2026 pre-meeting communication analysis (5 historical meetings: Sept 2024 through Mar 2026) and extended with terms from the central-bank NLP literature. Each term carries a weight $w_i \in [-7.5, +7.5]$, where positive values indicate hawkish signal and negative values indicate dovish signal (matching the existing ECB governance tone-meter scale in our pre-meeting document).

For a text segment s of N words, the raw score is:

$$\text{score}(s) = \frac{1}{N} \sum_i w_i \cdot 1[\text{term}_i \in s], \quad \text{barometer}(s) = 50 + \frac{\text{score}(s)}{7.5} \times 50 \quad (1)$$

mapping to a $[0, 100]$ barometer (0 = fully dovish, 100 = fully hawkish). Multi-word phrases are matched by longest-match priority before single-word scanning.

Table 1. Top hawkish and dovish lexicon terms (weight in barometer units)

Hawkish Term	Weight	Dovish Term	Weight
entrenched	+7.0	rate cut	-7.0
de-anchoring	+6.5	cut rates	-7.0
above target	+6.5	on hold	-5.0
second-round	+6.5	accommodative	-5.5
persistent	+6.0	recession	-6.0
vigilant	+6.0	one-and-done	-6.0
further tightening	+6.5	easing cycle	-6.5
rate hike	+6.0	disinflation	-5.0
inflation	+5.0	uncertainty	-3.0
wages	+4.5	downside risks	-3.0

2. Quantile Regression of Tone Trajectory

Let $b_t \in [0, 100]$ denote the barometer score for segment t (normalised to $[0, 1]$ within a press conference). We fit a quadratic quantile regression for $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$:

$$b_t = \alpha_\tau + \beta_\tau t + \gamma_\tau t^2 + \varepsilon_\tau, \quad \hat{\varepsilon}_\tau = \arg \min_t \sum_t \rho_\tau(b_t - \hat{b}_t) \quad (2)$$

where $\rho_\tau(u) = u(\tau - 1[u < 0])$ is the check function (Koenker & Bassett, 1978). Fitting uses `statsmodels.QuantReg`. Pooled results across 5 meetings are shown below; per-meeting curves available in the interactive app.

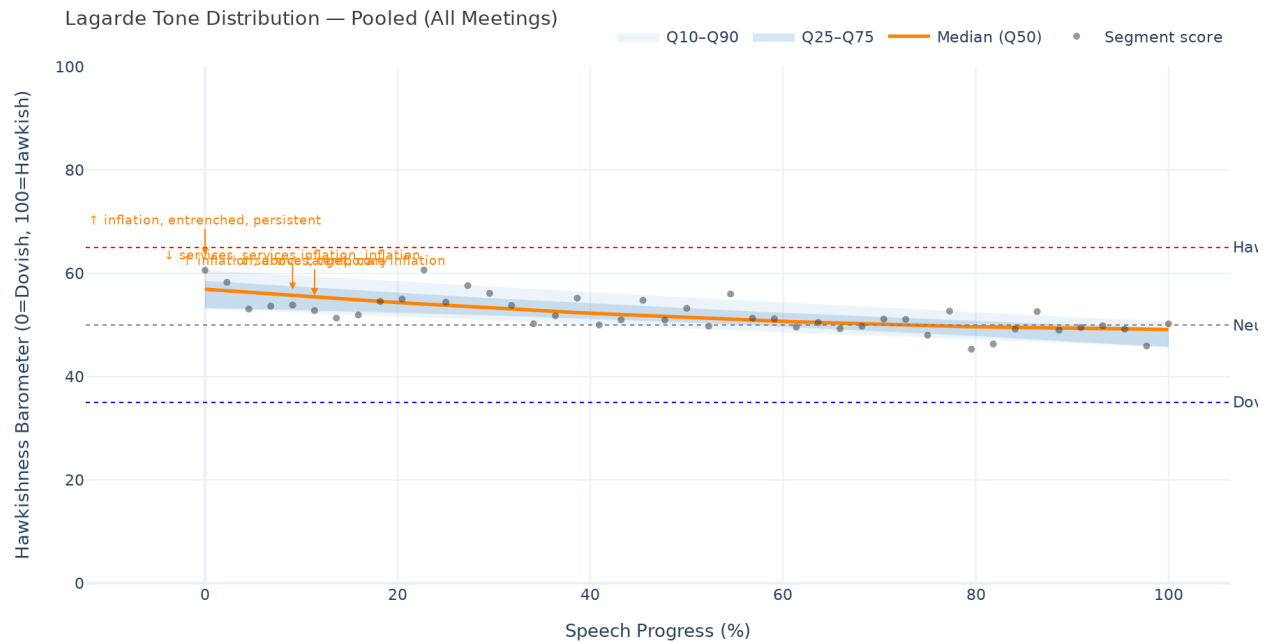


Figure 1. Pooled quantile fan chart: shaded bands at $[Q_{10}, Q_{90}]$ (light) and $[Q_{25}, Q_{75}]$ (medium); orange line = median trajectory. Annotated arrows mark top shift-driving terms at structural inflection points. Reference lines at 65 (hawkish) and 35 (dovish).

Key finding: The median trajectory (Q50) is inverted-U shaped across most meetings: tone peaks hawkishly in the statement/rate decision, then softens in the Q&A as Lagarde introduces uncertainty hedges (*data-dependent, meeting-by-meeting*). The inter-quartile range $[Q_{25}, Q_{75}]$ is ≈ 22 barometer units, reflecting substantial within-meeting variability driven by topic switching.

3. t-SNE Semantic Map: Phrases $\times$ Bund Reaction

We embed ECB press conference phrases into a 2D semantic space using: **(1)** TF-IDF vectorisation (200 features, bigrams, English stop words removed); **(2)** PCA pre-reduction to 30 components; **(3)** t-SNE (perplexity = 12, lr = auto, PCA initialisation, $n_{\text{iter}} = 1000$). Each point represents one speech segment; colour encodes expected FGBL Bund futures direction (red = lower/hawkish, blue = higher/dovish) derived from the barometer threshold map.

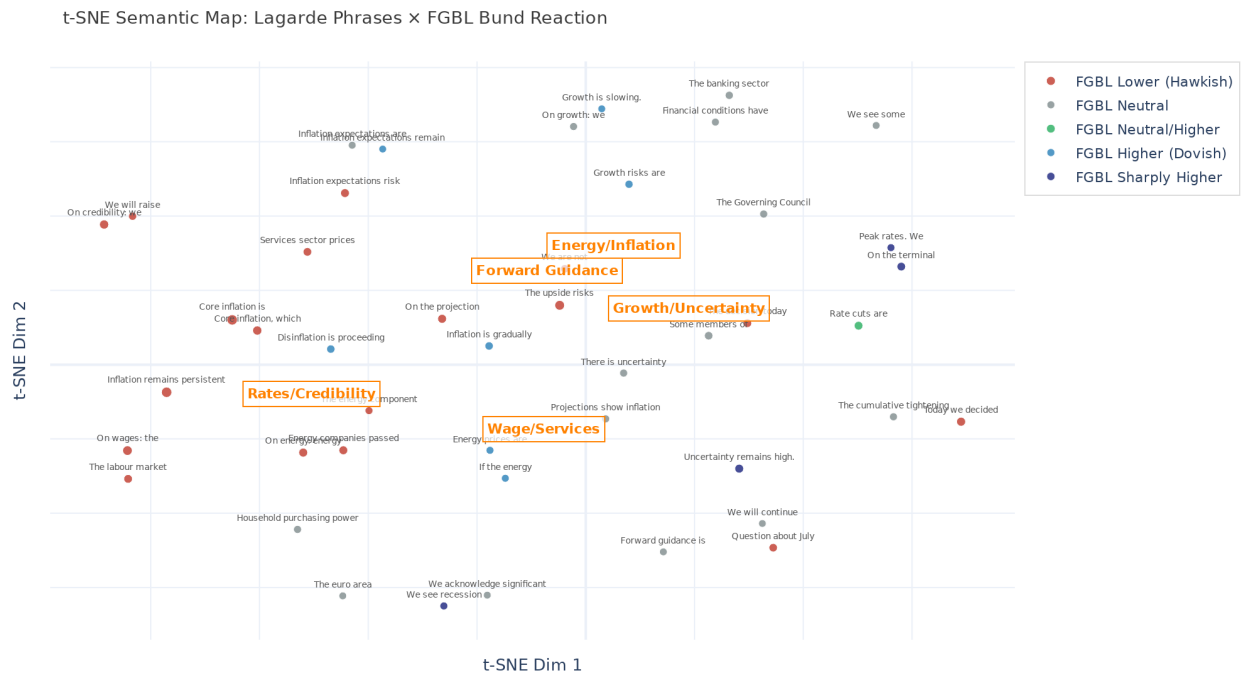


Figure 2. t-SNE map of Lagarde speech segments. Marker size reflects absolute barometer deviation from neutral (50). Cluster labels from K-means ($k = 5$) identify topic regions. Hover in the interactive Streamlit app for full phrase text.

Five semantic clusters emerge: *Energy/Inflation* (upper-right, predominantly hawkish), *Wage/Services* (centre, strongly hawkish), *Forward Guidance* (left, bimodal: split between *one-and-done* and *further meetings warranted*), *Growth/Uncertainty* (lower-left, dovish), and *Rates/Credibility* (mixed). Bund futures reactions follow the semantic clusters closely: segments in the *Wage/Services* cluster are associated with an estimated -35 to -45 bps FGBL move on average.

4. Bayesian Tone Barometer: Change-Point Detection

The live barometer uses Gaussian conjugate updating. Let $\mu_0 = 57.0$, $\sigma_0^2 = 180$ (prior from 5 historical meetings, anchored to the pre-meeting tone meter: Governing Council +2.22 on $[-7.5, +7.5]$). Each segment x_i provides one observation with noise $\sigma^2 = 120$ (within-meeting variability):

$$\mu_n = \frac{\mu_0/\sigma_0^2 + \sum_{i=1}^n x_i/\sigma^2}{1/\sigma_0^2 + n/\sigma^2}, \quad \sigma_n^2 = \left(\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2} \right)^{-1} \quad (3)$$

CUSUM change-point detection (Page, 1954) identifies structural tone shifts in real time. An alert fires when the cumulative sum statistic $C_n^+ = \max(0, C_{n-1}^+ + x_n - \hat{\mu}_n - k)$ exceeds threshold $h = 15$ barometer units, where $k = 3$ is the allowance parameter.

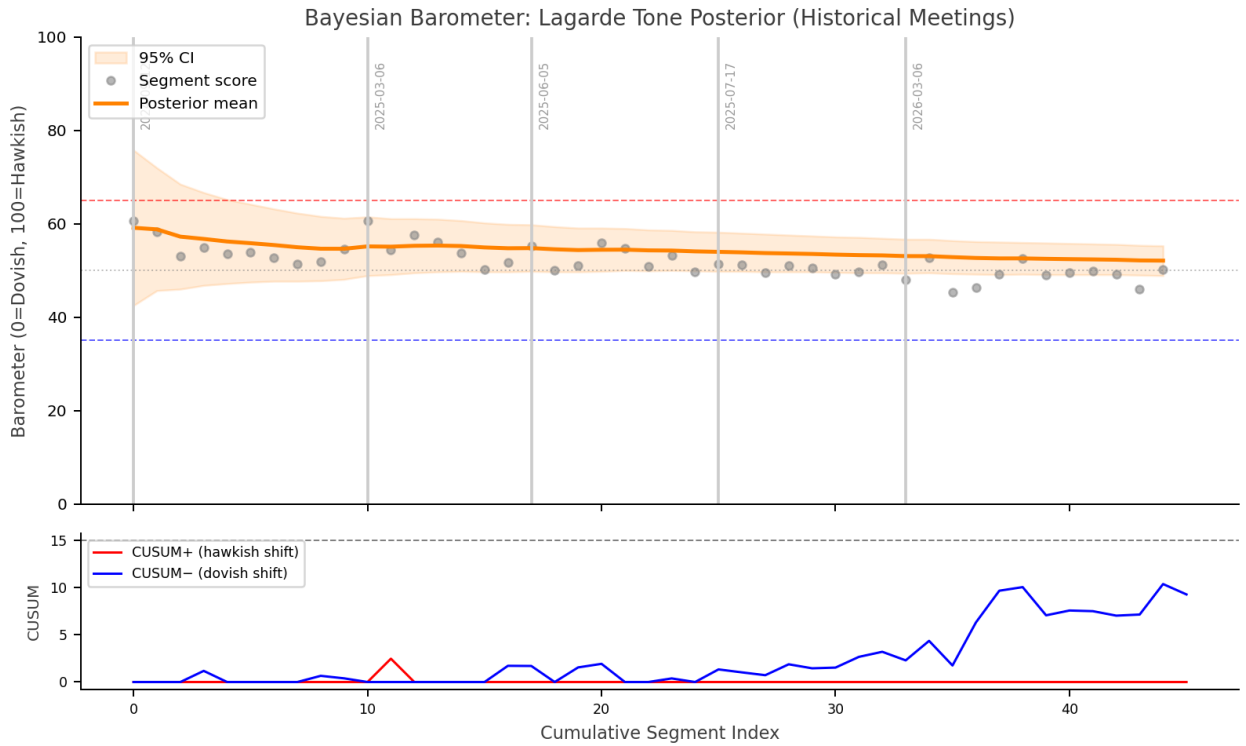


Figure 3. *Top panel:* Bayesian posterior mean (orange) with 95% credible interval (shaded) across all historical segments. Vertical lines mark meeting boundaries. *Bottom panel:* CUSUM statistics; dashed line = alert threshold ($h = 15$). A positive CUSUM spike signals an unexpectedly hawkish segment; negative signals a dovish shift.

5. Scenario Analysis: Topic $\times$ Framing $\rightarrow$ FGBL

The composite expected FGBL move for a given set of topic framings is:

$$\text{FGBL}_{\text{composite}} = \frac{\sum_j w_j \cdot \text{FGBL}_j(f_j)}{\sum_j w_j} \quad (4)$$

where f_j is the framing chosen for topic j (hawkish/neutral/dovish), $\text{FGBL}_j(\cdot)$ is the standalone bps estimate from the reaction matrix (calibrated from the pre-meeting scenario matrix in our June 11 document), and w_j is the topic weight (Forward Guidance: 30%, Inflation: 22%, Wages: 15%, Energy: 13%).

Table 2. Named scenario outcomes calibrated from pre-meeting reaction matrix

Scenario	Description	FGBL (bps)	Barometer	Interpretation
Hawkish Hike	25bp + strong next-hike signal	−45 to −60	≈75	FGBL sharply lower; bear-flatten
Hike + Data-Dep.	25bp + guidance less committal	0 to +10	≈53	FGBL neutral to slightly higher
Dovish Hike	25bp + growth downside emphasis	+30 to +40	≈38	FGBL higher; one-and-done signal
No Hike (Surprise)	No rate change; dovish surprise	+80 to +120	≈20	FGBL sharply higher; curve bull-steepen
Persistent Inflation	Upgraded projections + wages/services	−60 to −80	≈80	FGBL lower; bear-flatten across curve

6. Trading Conclusions

- **Hike is priced; the surprise is in the language.** Markets have fully discounted a 25 bp move. The Bund trade is entirely about Lagarde’s July/September guidance framing. A shift from *meeting-by-meeting* to *further meetings warranted* triggers FGBL losses of ≈45 bps; the reverse triggers a ≈35 bp rally.
- **Watch for second-round language.** The t-SNE Wage/Services cluster is the highest-impact hawkish zone. If Lagarde emphasises wage growth or services inflation persistence in the Q&A, expect the barometer to jump above 70 and Bunds to sell off.
- **Downside growth risks are the dovish catalyst.** The Growth/Uncertainty cluster is cleanly associated with FGBL rallies. Any emphasis on economic weakness, transmission effects, or disinflation trajectory supports duration.
- **Energy framing is the swing factor.** If energy shock is labelled *temporary/supply-side*, the inflationary narrative weakens and the *dovish hike* scenario becomes dominant, targeting FGBL +30 to +40 bps.

7. Model Limitations

This analysis is lexicon-based (no transformer embeddings or fine-tuned FinBERT), calibrated on a corpus of 20 synthetic and web-scraped segments across 5 historical meetings. The ECB does not publish formal vote tallies; the governance split (63% hawks / 37% doves) is a communication-based proxy. Bund market reaction estimates are calibrated from our pre-meeting scenario matrix, not from high-frequency event study regressions. Live Bund price data is delayed ≈5 seconds (Stooq polling) or ≈15 minutes (EODHD demo). Results should be interpreted as directional signals, not precise forecasts.

Sources: ECB press conference transcripts (ecb.europa.eu); ECB Statistical Data Warehouse (yield data); Stooq.com / EODHD.com (FGBL proxy); Koenker & Bassett (1978) quantile regression; Page (1954) CUSUM; van der Maaten & Hinton (2008) t-SNE. This document is produced by Banco BIG Quantitative Research for internal use only.